

HoloPresence: A Remote Monitoring System Using Autostereoscopic Display and Robot Arms

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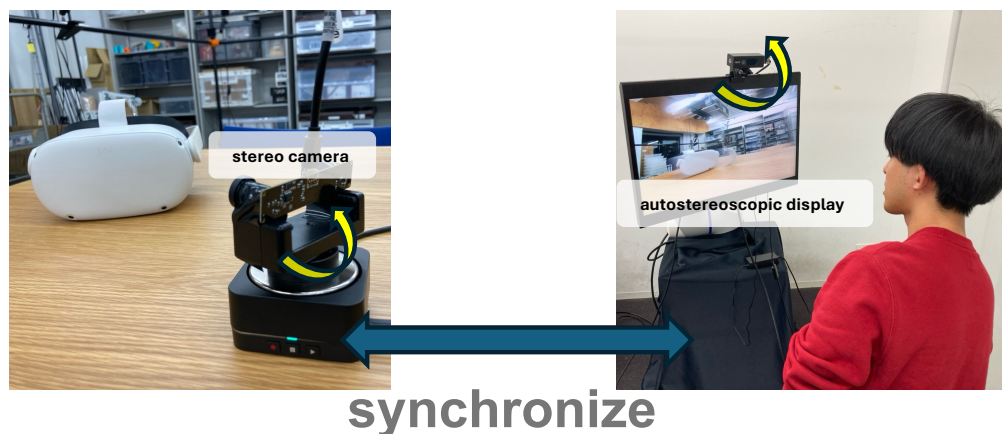


Figure 1: Our system, *HoloPresence*, is a three-dimensional remote monitoring system that dynamically presents remote images in response to user movements. When the user turns to the left, the autostereoscopic display and stereo cameras mounted on two robot arms pivot to the left, maintaining alignment with the user's viewpoint.

ABSTRACT

Remote monitoring is widely used in various fields as it allows users to grasp remote scenes more intuitively. Conventional user interfaces mainly rely on 2D displays, while recent advances in 3D display technology have prompted the exploration of HMD-based solutions. However, although HMDs offer depth perception, they also introduce issues such as cumbersome headgear and VR sickness. Therefore, we propose a novel remote monitoring system called *HoloPresence*, which employs an autostereoscopic display and a stereo camera—each mounted on a separate robot arm—and dynamically controls their orientations according to the user's viewpoint. This setup allows users to observe remote scenes in 3D with a wide viewing angle, without the need for additional wearable devices. In this paper, we describe our prototype implementation of *HoloPresence* and present the findings from a pilot study conducted to qualitatively evaluate the system.

Index Terms: Human-centered computing Visualization; Human-centered computing Mixed / augmented reality

1 INTRODUCTION AND BACKGROUND

Remote monitoring enables real-time observation of distant locations and is widely used in fields such as industry, disaster response,

healthcare, and nursing care[1]. Its primary advantage is the ability to safely oversee hazardous environments (e.g., factories) without requiring physical presence. To achieve this, systems with cameras mounted on drones[2] and robots[3] have been developed.

Although considerable research has focused on autonomous camera movements, the user interface and display technologies have not been explored in depth. Most remote monitoring systems still rely on 2D displays, which lack sufficient depth information and make it difficult to accurately perceive spatial relationships[4]. This limitation is critical because depth perception is essential for effective remote monitoring.

To address the lack of depth information, Yun et al.[5] proposed using head-mounted displays (HMDs) for video presentation and demonstrated their effectiveness. However, HMDs also present several drawbacks, including discomfort, VR sickness, and fatigue during prolonged use[6]. Additionally, if users are not continuously wearing the device, they risk missing important events. Although HMDs provide valuable 3D depth information, these issues hinder their widespread adoption as remote monitoring interfaces. Hence, an alternative 3D display solution is needed.

Autostereoscopic displays are one such solution, as they offer 3D viewing without requiring additional equipment[7]. They are generally more comfortable and less likely to cause physical discomfort than HMDs[6], making them a promising option for remote monitoring. However, current autostereoscopic displays suffer from narrow viewing angles[8], limiting their effectiveness in delivering comprehensive spatial information.

To overcome these challenges, we propose a new remote monitoring system that enables 3D viewing of distant locations with a wide field of view using an autostereoscopic display (Fig. 1). Our system integrates Dendo et al.'s *HoloArm*[9] with a robot arm equipped with a stereo camera. By coordinating their movements based on the

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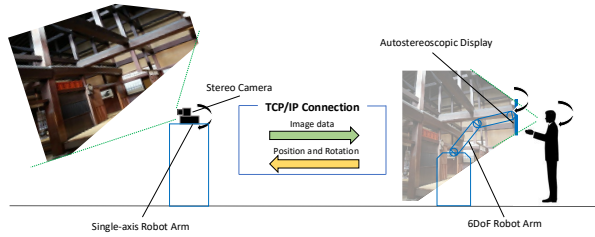


Figure 2: A schematic illustration of system overview. The autostereoscopic display is mounted on a 6DoF robot arm (right) and the stereo camera on a 1-axis arm (left). Both are communicating via a TCP/IP connection, with the stereo camera transmitting images from the remote location and the robot arm equipped with the autostereoscopic display transmitting its own position and rotation.

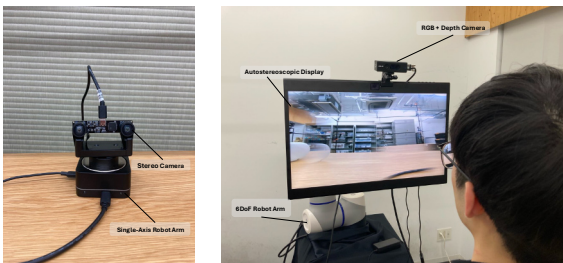


Figure 3: This is an actual photograph of the HoloPresence. The component installed in a remote location and takes pictures of the remote location (left) uses a stereo camera and a 1-axis arm. The component that presents images to the user (right) uses an autostereoscopic display, a 6DoF robot arm, and an RGB+Depth camera to capture the user's head movement.

user's viewpoint, the two robot arms allow users to perceive detailed depth information from a broad range of angles—all without the need for an HMD.

This paper describes the design and implementation of our prototype system. We also conducted a pilot study aimed at qualitatively investigating the system's usability and performance.

2 SYSTEM OVERVIEW

Fig. 2 shows the system overview. This system consists of two robot arms that communicate with each other via a TCP/IP connection. One robot arm is equipped with an autostereoscopic display, while the other carries a stereo camera. The stereo camera captures images in real-time and transmits them to the autostereoscopic display, where they are projected in three dimensions.

The system operates as follows. First, the user observes the remote location through the autostereoscopic display. When the user turns their head to the left (or otherwise shifts their viewpoint) beyond the display's initial viewing angle, the 6DoF robot arm rotates to reposition the display. The rotation and position information are then sent to the 1-axis arm, which rotates the stereo camera accordingly. This coordination between the two arms allows the user to view a remote location from a wider angle of view than conventional autostereoscopic displays can provide.

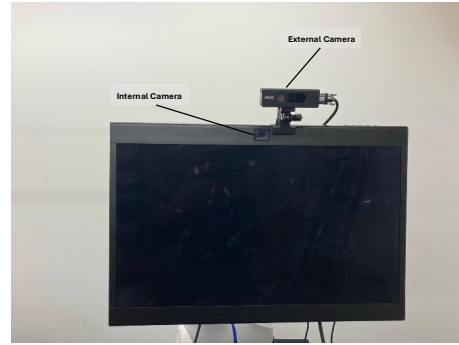


Figure 4: Two cameras follow the user. The external camera captures the position and rotation of the human head and transmits its position to the robot arm. The internal camera captures the position and rotation of the human eye and renders a three-dimensional image accordingly at high speed.

3 IMPLEMENTATION

3.1 Apparatus

Fig. 3 shows an actual photograph of the HoloPresence system. We used the Sony Elf SR2 as an autostereoscopic display to present images to the user. Because this display weighs 8 kg, we adopted a Dobot CR10 robot arm (payload: 10 kg) to support it. To enable the robot arm to follow the user, we attached an ASUS Xtion2 (RGB+Depth) camera. Additionally, we employed an ELPCAM USB stereo camera to capture the remote location. This stereo camera acquires separate RGB images for the left and right eyes, and it is mounted on a single-axis robot arm (KeiganMotor).

Since the primary goal of this project was to demonstrate the feasibility of a remote monitoring system with a wide viewing angle via an autostereoscopic display, we opted for a single-axis robot arm for simplicity. We deemed this sufficient to extend the viewing angle of the autostereoscopic display while reducing overall system complexity.

3.2 Capturing and Following User

As illustrated in Fig. 4, the system employs two distinct mechanisms to track the user. One is an external camera attached to the display. We use OpenSeeFace[10], developed by Emiliania, to capture the translation and rotation of the user's head. The robot arm then moves according to this head-position data.

The second mechanism is an internal camera built into the ELF SR2 (autostereoscopic display). This internal camera tracks the user's eyes and renders a 3D image in real-time. By combining these two cameras—an external one that tracks the user's head and an internal one that tracks the user's eyes—the system maintains robust, uninterrupted tracking.

Moreover, as long as the user's head remains within the internal camera's tracking range, that camera can continue to track and render the 3D view at high speed, independently of the robot arm. This means the internal camera can update the display before the robot arm finishes repositioning itself, thereby minimizing any perceived delay from the user's perspective.

3.3 Controlling Robot

We used Unity to control the Dobot CR10 and Python to control the Keigan motor. The two robot arms communicate via a TCP/IP connection, and the Keigan motor constantly acquires the Dobot CR10's position and rotation data. When Dobot CR10 follows the user, we apply linear completion and smoothing mechanisms. Specifically, linear completion moves the arm to 50% of the target

position each frame, and smoothing is achieved by taking the average of the previous five frames as the target value.

This approach ensures the system can follow the user's movements smoothly while filtering out sudden outliers caused by tracking errors, thereby improving both stability and safety.

4 USER FEEDBACK AND DISCUSSION

Feedback was collected from 4 users who tested the prototype. Their comments covered various aspects, including the accuracy and speed of tracking, as well as potential interactive features. Several participants noted that the internal camera's real-time response to their movements, in addition to the robot arm's tracking, enhanced their sense of immersion. One participant suggested adding a hand interaction, enabling users to feel like they are looking through a virtual window into the remote location. Another participant proposed applications in entertainment—such as sports viewing or live music—where the system could automatically adjust the viewpoint to the user's preferred angle.

On the other hand, a few participants mentioned that the arm's tracking was not fully real-time and introduced noticeable delays. Others reported eye strain when looking directly at the 3D images. To address these concerns, we plan to implement predictive tracking to move the robot arm almost simultaneously with the user, as well as adaptive stereoscopic rendering that adjusts to the user's interpupillary distance.

5 CONCLUSION AND FUTURE WORK

This paper introduces HoloPresence, an innovative remote monitoring system that integrates an autostereoscopic display with two robot arms. Unlike traditional 2D displays, HoloPresence leverages an autostereoscopic display to provide depth perception, offering a more intuitive monitoring experience compared to head-mounted displays (HMDs). Additionally, by incorporating robot arms, HoloPresence overcomes a significant limitation of conventional autostereoscopic displays—the narrow viewing angle—thereby enhancing usability. In future work, we plan to validate the system's effectiveness through extensive user experiments and further improve its capabilities by utilizing a 6-axis robot arm, such as the xArm, for more precise camera mounting.

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